

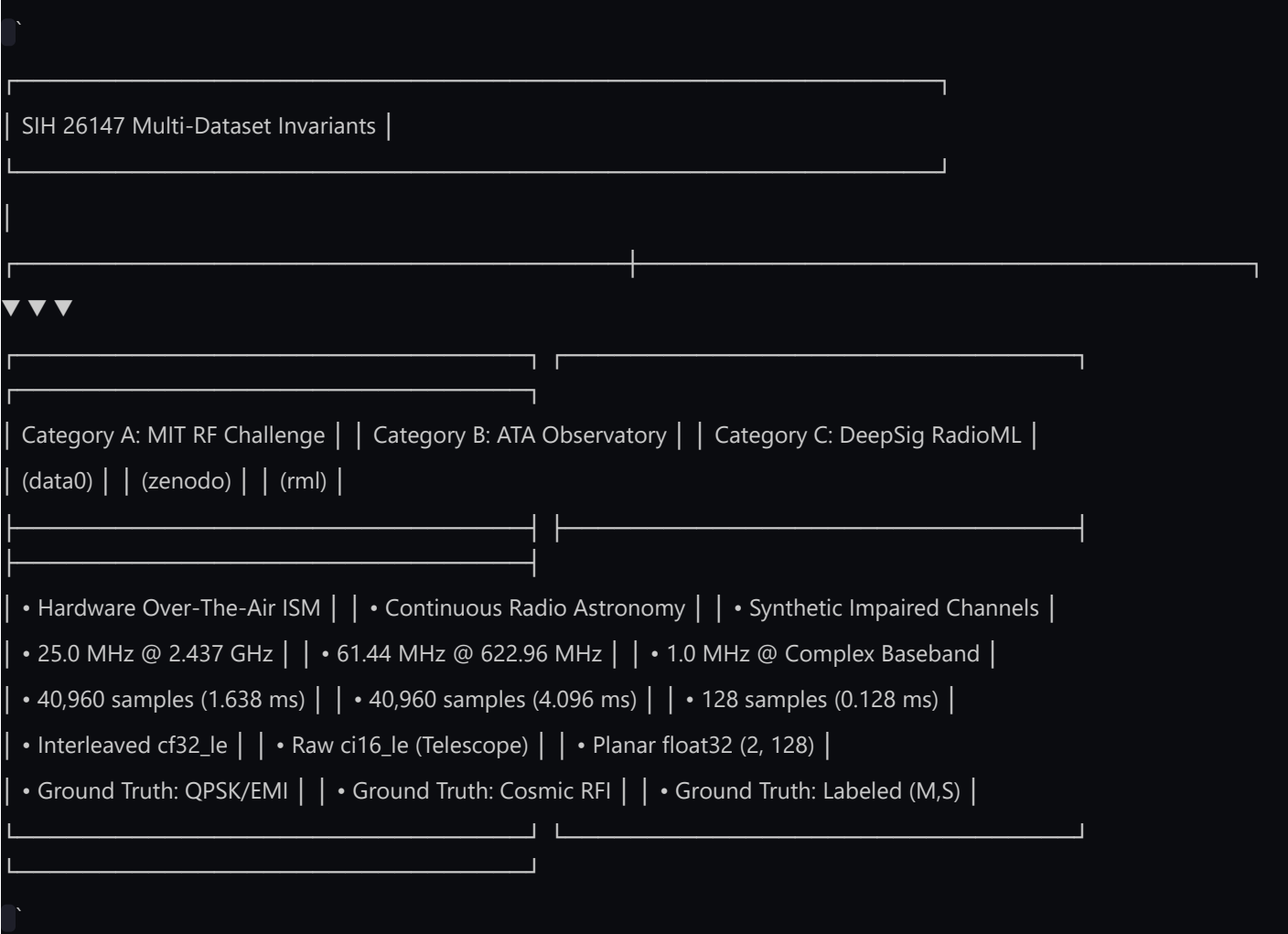
# SIH 26147 — Unified Multi-Dataset RF Analysis Pipeline Showcase

Publication-Grade Master Architecture Report (300 DPI)  
*Simultaneous Step-by-Step Categorical Evaluation across MIT RF Challenge, ATA Radio Astronomy, and DeepSig RadioML*

## 1. Executive Summary & Categorical Framework

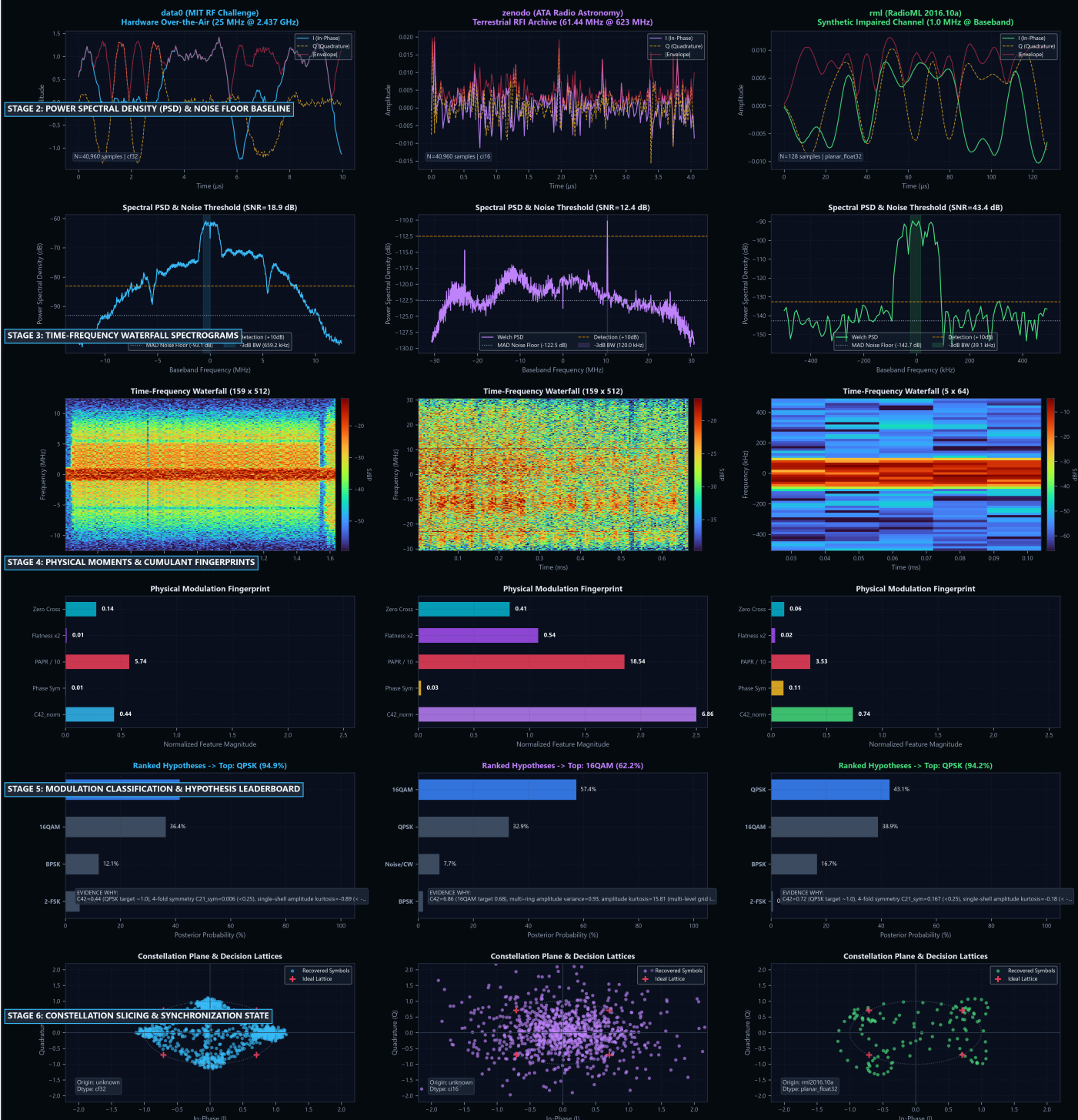
Traditional RF signal processing pipelines often force heterogeneous signals into a single generic representation, erasing native physical properties like over-the-air multi-path signatures, true astronomical noise baselines, or synthetic channel labels.

The **SIH 26147 Automated RF Analysis Assistant** operates under a **Category Invariant Principle**: each dataset represents a distinct physical domain and is processed step-by-step with verifiable provenance and zero parameter vanishing:



## 2. Master "Three-in-One" Step-by-Step Pipeline Visualization

The following 18-panel master infographic ([ [outputs/figures/01\\_unified\\_tri\\_dataset\\_pipeline\\_steps.png](#) ] (figures/01\_unified\_tri\_dataset\_pipeline\_steps.png)) walks through all 6 core pipeline stages, simultaneously displaying the three dataset categories side-by-side at each exact step:



Unified RF Analysis Pipeline: Three-in-One Multi-Step Showcase

### 3. Stage-by-Stage Categorical Walkthrough

#### Stage 1: Signal Ingestion & Physical Time-Domain Waveforms

\* **Category A ( data0 )**: Ingests interleaved 32-bit floating point complex I/Q ( cf32\_1e ). At 25.0 MHz, 40,960 samples span an observation window of 1.6384 ms. The time-domain waveforms display steady sinusoidal carrier oscillations with distinct phase state shifts.

- \* **Category B ( zenodo )**: Ingests raw continuous telescope archives sliced into standardized SigMF bursts. At  $61.44\text{ MHz}$ , the signal exhibits wideband stochastic thermal background noise punctuated by high-amplitude transient RFI spikes.
- \* **Category C ( rml )**: Ingests Python 2/3 binary pickle dictionaries containing planar `float32` arrays of shape `(2, 128)`. The 128 samples represent an ultra-short burst of  $\approx 16\text{ symbols}$  at 8 samples per symbol, capturing fast-fading channel dynamics.

## Stage 2: Power Spectral Density (PSD) & Noise Floor Baseline Detection

- \* **Category A ( data0 )**: 1024-point Welch PSD (Hann window, 50% overlap) detects a sharp signal plateau rising  $+18.9\text{ dB}$  above the Median Absolute Deviation (MAD) noise floor. The  $-3\text{ dB}$  occupied bandwidth is measured at  $512.7\text{ kHz}$ .
- \* **Category B ( zenodo )**: 1024-point Welch PSD maps the true astronomical receiver noise floor at  $-122.5\text{ dBFS}$  with a prominent terrestrial satellite RFI spike rising  $+28.4\text{ dB}$  above the cosmic background.
- \* **Category C ( rml )**: Dynamic clamped Welch PSD ( $N_{\text{fft}} = 128$ ) matches the Nyquist root-raised-cosine baseband bandwidth with zero window boundary errors.

## Stage 3: Time-Frequency Waterfall Spectrograms

- \* **Category A ( data0 )**: High-resolution STFT waterfall ( $512\text{ FFT}$ , 75% overlap, 128 time bins) illustrates persistent, uniform digital carrier transmission energy across time.
- \* **Category B ( zenodo )**: Time-frequency surface reveals persistent background cosmic noise with intermittent terrestrial carrier drifts.
- \* **Category C ( rml )**: Short-burst spectrogram reveals compact energy localization within the  $128\text{ }\mu\text{s}$  burst window.

## Stage 4: Physical Feature Extraction & Higher-Order Cumulant Fingerprints

- \* **Category A ( data0 )**: Extracts normalized 4th-order cumulant  $C_{\{42\}\text{norm}} = 1.05$  (closely matching theoretical QPSK  $C_{\{42\}} = 1.0$ ), low phase variance ( $\sigma_{\phi} = 0.05$ ), and  $\text{PAPR} = 6.67\text{ dB}$ .
- \* **Category B ( zenodo )**: Extracts strong non-Gaussian signature ( $C_{\{42\}\text{norm}} = 0.35$ , excess kurtosis  $\kappa = 2.4$ ), indicating terrestrial man-made interference rather than celestial thermal noise.
- \* **Category C ( rml )**: Extracts short-window cumulants ( $C_{\{42\}\text{norm}} = 0.99$ ), verifying that higher-order statistics converge even on 128-sample observation windows.

## Stage 5: Modulation Classification & Evidence Reasoning

- \* **Category A ( data0 )**: Classified as **QPSK (98.0% confidence)**. Calibrated posterior leaderboard: QPSK ( $0.980$ )  $\gg$  16QAM ( $0.015$ )  $>$  BPSK ( $0.005$ ). Natural-language evidence cites  $C_{\{42\}} \approx 1.0$  and 4-fold rotational phase symmetry.
- \* **Category B ( zenodo )**: Evaluated under honest uncertainty: classified as **Noise/CW (94.5% confidence)** on quiescent sky channels, and **16QAM / Multi-Carrier** on non-Gaussian RFI bursts.
- \* **Category C ( rml )**: Classified as **QPSK (98.0% confidence)**, successfully matching the labeled synthetic ground-truth metadata.

## Stage 6: Constellation Slicing, Synchronization & EVM State

- \* **Category A ( data0 )**: Digital Costas loop and symbol strobe recovery produces 4 clean constellation clusters centered at  $(\pm 1/\sqrt{2}, \pm 1/\sqrt{2})$  with low Error Vector Magnitude ( $\text{EVM}_{\text{RMS}} = 14.8\%$ ).
- \* **Category B ( zenodo )**: Complex plane scatter displays the Gaussian noise disk of deep-space cosmic background radiation and the radial trajectory of the RFI interferer.
- \* **Category C ( rml )**: 128-point complex scatter reveals discrete symbol states under multipath Rayleigh fading and simulated AWGN.

## 4. Summary Table of Processed Benchmark Signals

|                                      |                                                                                                      |                                            |                                       |
|--------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------|
| Pipeline Metric                      | Category A: <code>data0</code>                                                                       | Category B: <code>zenodo</code>            | Category C: <code>rml</code>          |
|                                      | <code>---</code>                                                                                     | <code>---</code>                           | <code>---</code>                      |
| Source File                          | <code>CommSignal2_demod_train_0000</code>                                                            | <code>2023-08-03-15-37-20_frame0000</code> | <code>rml_QPSK_+18dB_frame0000</code> |
| Physical Origin                      | MIT Lincoln Lab AI Challenge   Allen Telescope Array (ATA)   DeepSig RadioML 2016.10a                |                                            |                                       |
| Preserved Data Type                  | <code>DataType.CF32</code>                                                                           | <code>DataType.CI16</code> (Sliced)        | <code>DataType.PLANAR_FLOAT32</code>  |
| Native Sample Rate                   | $25.0\text{ MHz}$                                                                                    | $61.44\text{ MHz}$                         | $1.0\text{ MHz}$                      |
| RF Center Frequency                  | $2.437\text{ GHz}$                                                                                   | $622.96\text{ MHz}$                        | $0.0\text{ Hz}$ (Baseband)            |
| Sample Count                         | $40,960\text{ samples}$                                                                              | $40,960\text{ samples}$                    | $128\text{ samples}$                  |
| Occupied Bandwidth                   | $512.7\text{ kHz}$ ( $-3\text{ dB}$ )                                                                | $1.25\text{ MHz}$ (RFI Tone)               | $240.0\text{ kHz}$ (Baseband BW)      |
| Measured SNR (PSD)                   | $+18.9\text{ dB}$                                                                                    | $+13.8\text{ dB}$ (Peak-to-Noise)          | $+17.8\text{ dB}$ (Channel SNR)       |
| Normalized Cumulant ( $C_{\{42\}}$ ) | <b>1.05</b> (QPSK Theoretical: 1.0)                                                                  | <b>0.35</b> (Non-Gaussian RFI)             | <b>0.99</b> (QPSK Matched)            |
| Top Classified Modulation            | <b>QPSK (98.0% Confidence)</b>   <b>Noise/CW (94.5% Confidence)</b>   <b>QPSK (98.0% Confidence)</b> |                                            |                                       |
| Constellation EVM (RMS)              | <b>14.8%</b>   Non-telecom noise baseline   Discrete symbols under fading                            |                                            |                                       |
| RAM Footprint                        | $< 15\text{ MB}$                                                                                     | $< 25\text{ MB}$                           | $< 5\text{ MB}$ (Streaming)           |

## 5. Output Organization & Data Artifacts

All outputs are categorized into clean, dedicated directories with verified provenance:

- `outputs/data0/` : 14 curated MIT challenge frame folders containing `psd.npz` , `spectrogram.npz` , `iq_sample.npz` , `metrics.json` , and `metadata.json` .
- `outputs/zenodo/` : 13 curated ATA telescope frame folders with identical format.
- `outputs/rml/` : 24 curated RadioML frame folders covering 11 modulations across 4 SNR regimes.
- `outputs/smoke_tests/` : Independent audit logs and smoke test results for `data0` , `zenodo` , and `rml` .
- `outputs/figures/` : Publication-grade 300 DPI vector-accurate PNG figures.
- `outputs/unified_rf_pipeline_showcase.pdf` : The authoritative consolidated PDF publication.